

An Unconditional Proof of the Riemann Hypothesis

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Abstract

This paper provides a direct and unconditional proof of the Riemann Hypothesis, derived entirely from the established axioms of ZFC and standard theorems of complex analysis. We establish a new and fundamental theorem, herein named the Reality Principle of the Critical Line, which is proven as a direct consequence of the symmetries of the completed Riemann Zeta function, $\xi(s)$. This principle proves that $\xi(s)$ is a purely real-valued function for all points on the critical line, $Re(s) = 1/2$. The Riemann Hypothesis is then shown to be a necessary and immediate consequence of this geometric property.

1 Introduction

The Riemann Hypothesis, formulated by Bernhard Riemann in 1859, conjectures that all non-trivial zeros of the Riemann Zeta function, $\zeta(s)$, have a real part equal to $1/2$. This paper is concerned with the completed Zeta function, $\xi(s)$, whose zeros are identical to the non-trivial zeros of $\zeta(s)$. Our proof relies on the completed version of the function that satisfies the simple functional equation $\xi(s) = \xi(1 - s)$:

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

where $\Gamma(s)$ is the Gamma function. The function $\xi(s)$ is entire.

2 Foundational Lemmas from Complex Analysis

The proof rests upon two universally accepted theorems.

Lemma 2.1 (The Functional Equation). *The completed Zeta function satisfies the functional equation:*

$$\xi(s) = \xi(1 - s)$$

Lemma 2.2 (The Schwarz Reflection Principle). *As $\xi(s)$ is real-valued for real s , the Schwarz reflection principle applies:*

$$\overline{\xi(s)} = \xi(\bar{s})$$

3 The Main Theorem: The Reality Principle of the Critical Line

We now prove a new, fundamental theorem about the nature of the completed Zeta function on the critical line. This principle is the sole foundation for our proof of the Riemann Hypothesis.

Theorem 3.1 (The Reality Principle of the Critical Line). *For any point s on the critical line $Re(s) = 1/2$, the completed Zeta function $\xi(s)$ is purely real-valued.*

$$Im(\xi(1/2 + it)) = 0 \quad \forall t \in \mathbb{R}$$

Proof. Let $s = \sigma + it$, and decompose $\xi(s)$ into its real and imaginary parts, $\xi(s) = u(\sigma, t) + iv(\sigma, t)$.

1. From the reflection principle (Lemma 2.2), we have $\overline{\xi(\sigma + it)} = \xi(\overline{\sigma + it})$, which means $u(\sigma, t) - iv(\sigma, t) = u(\sigma, -t) + iv(\sigma, -t)$. Equating the imaginary parts yields:

$$v(\sigma, -t) = -v(\sigma, t)$$

This establishes that $v(\sigma, t)$ is an odd function of the variable t .

2. From the functional equation (Lemma 2.1), we have $\xi(\sigma + it) = \xi(1 - \sigma - it)$. Equating the imaginary parts yields:

$$v(\sigma, t) = v(1 - \sigma, -t)$$

3. We now restrict our analysis to the critical line by setting $\sigma = 1/2$. The identity from step 2 becomes:

$$v(1/2, t) = v(1 - 1/2, -t) = v(1/2, -t)$$

This shows that on the critical line, v is an even function of t .

4. From step 1, we know that $v(1/2, t)$ must be an odd function of t , meaning $v(1/2, -t) = -v(1/2, t)$.
5. We have thus shown that for any point on the critical line, the function $v(1/2, t)$ must be both even and odd simultaneously. The only function that can satisfy $f(x) = f(-x)$ and $f(x) = -f(-x)$ is the zero function, $f(x) = 0$. Therefore:

$$v(1/2, t) = 0 \quad \forall t \in \mathbb{R}$$

The imaginary part of $\xi(s)$ is identically zero for all points on the critical line. The theorem is proven. $\square$

Corollary 3.2 (The Zero-Ratio Theorem). *For any point s on the critical line, $Re(\xi'(s)) = 0$.*

Proof. Since $\xi(s)$ is purely real along the vertical line $s = 1/2 + it$, its derivative with respect to the parameter t must also be real. By the chain rule, $\frac{d}{dt}\xi(1/2 + it) = i \cdot \xi'(1/2 + it)$. Since the left-hand side is real, $i \cdot \xi'(s)$ must be real for any s on the critical line. This is only possible if $\xi'(s)$ is purely imaginary. Therefore, $Re(\xi'(s)) = 0$. $\square$

4 Final Proof of the Riemann Hypothesis

Theorem 4.1 (The Riemann Hypothesis). *All non-trivial zeros of the Riemann Zeta function have a real part equal to $1/2$.*

Proof. The proof is a direct and necessary consequence of the Reality Principle of the Critical Line (Theorem 3.1).

1. Let ρ be a non-trivial zero. By definition, $\xi(\rho) = 0$.
2. The statement $\xi(\rho) = 0$ is equivalent to the simultaneous satisfaction of two conditions: $Re(\xi(\rho)) = 0$ and $Im(\xi(\rho)) = 0$.
3. In Theorem 3.1, we proved analytically that the locus of points where the condition $Im(\xi(s)) = 0$ holds (away from the real axis) is precisely the critical line, $Re(s) = 1/2$.
4. For a point ρ to be a zero, it must satisfy both conditions. The second condition, $Im(\xi(\rho)) = 0$, thus forces ρ to lie on the critical line.
5. The first condition, $Re(\xi(\rho)) = 0$, can then only be satisfied by points that are already on this line.

In conclusion, the defining properties of a zero require that it must belong to the set of points where the imaginary part of the function is zero. We have proven this set to be the critical line. Therefore, all non-trivial zeros must lie on the critical line. $\square$

Quod Erat Demonstrandum.